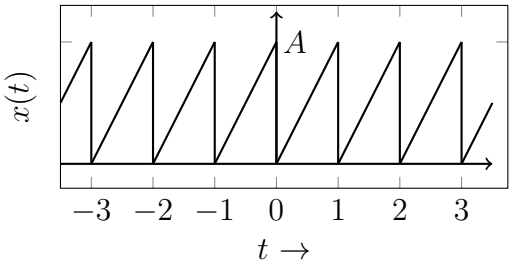
EE1101 Signals and Systems Jan—May 2021
Tutorial 5 Solutions

1) (a) The signal $x(t)$ is periodic with period $T_0 = 1$ and the fundamental frequency $f_0 = \frac{1}{T_0} = 1$ Hz, and $\omega_0 = \frac{2\pi}{T_0} = 2\pi$.



The Fourier series representation of the signal is,

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

where a_k's are the Fourier coefficients given by

$$\begin{aligned} a_k &= \frac{1}{T_0}\int_{T_0} x(t)e^{-jk\omega_0 t}dt \\ &= A\int_0^1 te^{-jk2\pi t}dt \\ &= \frac{Aj}{2\pi k} \qquad \text{for } k \neq 0 \\ a_0 &= \frac{1}{T_0}\int_{T_0} x(t)dt = \frac{A}{2} \end{aligned}$$

The magnitude of Fourier coefficient $|a_k| = \frac{A}{2\pi k}, k \neq 0$ and $|a_0| = \frac{A}{2}$. The phase $\arg(a_k) = sign(k)j$ i.e.,

$$\arg(a_k) = \begin{cases} \pi/2 & \text{if } k > 0 \\ -\pi/2 & \text{if } k < 0 \\ 0 & \text{if } k = 0 \end{cases}$$

The magnitude and phase spectrum are presented in Fig. 2. For these plots we assume $A = 1$.

And the Fourier series representation of the signal is

$$\begin{aligned} x(t) &= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \\ &= a_0 + \sum_{k\neq 0} a_k e^{jk\omega_0 t} \\ &= \frac{A}{2} + \sum_{k\neq 0} \frac{A}{2\pi k} e^{j(k\omega_0 t + \frac{\pi}{2})} \end{aligned}$$

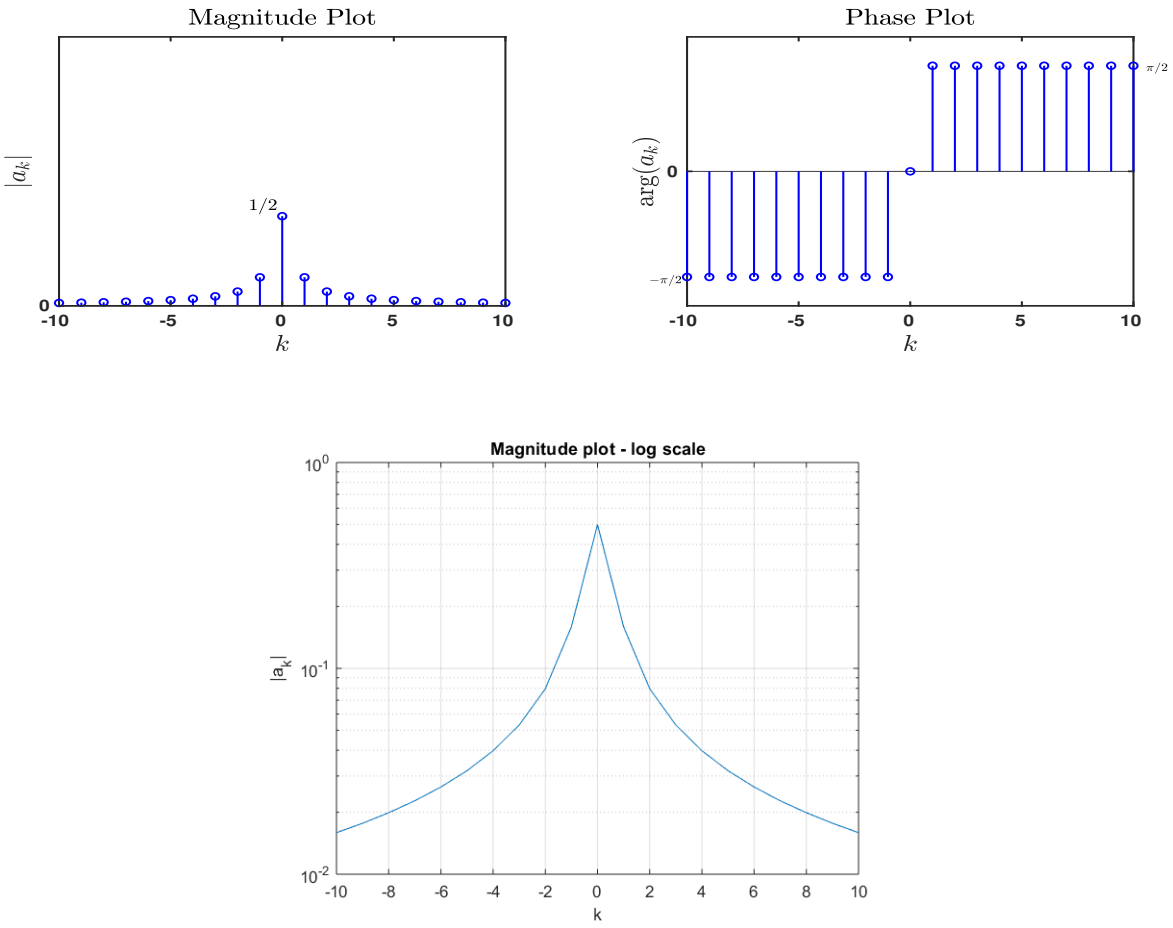


Figure 1: Q1.a) Magnitude and phase spectra of $x(t)$ for $A = 1$.

(b) The period of function $y(t)$ is $T_0 = 1$ and the fundamental frequency $\omega_0 = \frac{2\pi}{T_0} = 2\pi$. Let b_k indicate the Fourier coefficients of the signal $y(t)$. The Fourier series representation of the signal is,

$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega_0 t}$$

where b_k's are given by

$$b_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

Consider the period $-1/2 \leq t \leq 1/2$, $y(t) = -At + \frac{3A}{2}$. Computing the Fourier coefficients,

$$\begin{aligned} b_k &= \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt \\ &= \int_{-.5}^{.5} (-At + \frac{3A}{2}) e^{-jk\omega_0 t} dt \\ &= -A \int_{-.5}^{.5} t e^{-j2\pi kt} dt + \frac{3A}{2} \int_{-.5}^{.5} e^{-j2\pi kt} dt \end{aligned}$$

The second term integrates to zero. Using integration by parts on first term ($\int udv = uv - \int vdu$),

$$\begin{aligned} \int_{-.5}^{.5} t e^{-j2\pi kt} dt &= \frac{j}{2\pi k} cos(\pi k) \\ &= \frac{j}{2\pi k} e^{-j\pi k} \end{aligned}$$

Therefore,

$$b_k = \frac{-jA}{2\pi k} e^{-j\pi k} \quad k \neq 0$$

$$\begin{aligned} b_0 &= \frac{1}{T_0} \int_{T_0} x(t) dt \\ &= \int_{-.5}^{.5} (-At + \frac{3A}{2}) dt \\ &= -\frac{-At^2}{2} + \frac{3A}{2} t |_{-.5}^{.5} \\ &= \frac{3A}{2} \end{aligned}$$

The magnitude of Fourier coefficient $|b_k| = \frac{A}{2\pi k}, k \neq 0$ and $|b_0| = \frac{3A}{2}$. The phase $\arg(b_k)$ for $k \geq 0$ is given by

$$\arg(b_k) = \begin{cases} \pi/2 & \text{if } k > 0 \text{ is odd} \\ -\pi/2 & \text{if } k > 0 \text{ is even} \\ 0 & \text{if } k = 0 \end{cases}$$

Since the signal is real, the phase spectrum should be antisymmetric. Thus, the phase for $k < 0$ is

$$\arg(b_k) = \begin{cases} \pi/2 & \text{if } k < 0 \text{ is even} \\ -\pi/2 & \text{if } k < 0 \text{ is odd} \\ 0 & \text{if } k = 0 \end{cases}$$

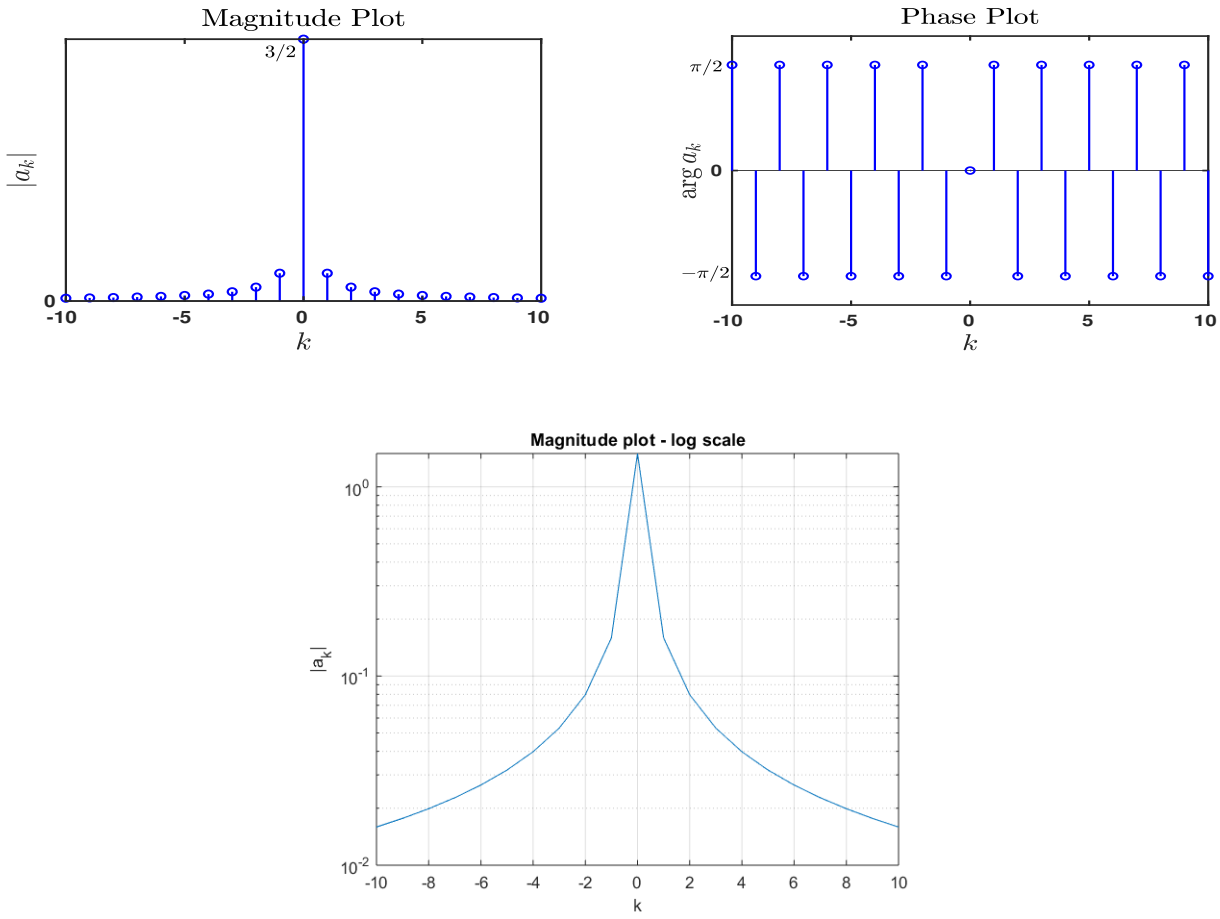


Figure 2: Q1.b) Magnitude and phase spectra of $y(t)$ for $A = 1$.

The magnitude and phase spectra are presented in Fig. 3. For these plots we assume $A = 1$.

(c) The given function is $y(t) = A + x(-t + 0.5)$.
Fourier coefficients of

$$\begin{aligned} x(t) &\longrightarrow a_k \\ x(t + 0.5) &\longrightarrow e^{jk\pi} a_k \\ x(-t + 0.5) &\longrightarrow e^{-jk\pi} a_{-k} \end{aligned}$$

Therefore, Fourier coefficients of $y(t)$

$$\begin{aligned} b_k &= \frac{-Aj}{2\pi k} e^{-jk\pi} \\ b_0 &= \frac{3A}{2} \end{aligned}$$

2) (a) The plot of signal $x_1(t)$ is shown below,

$$x_1(t) = \begin{cases} 1 & |t| < \frac{1}{4} \\ 0 & \frac{1}{4} < |t| < \frac{1}{2} \end{cases}$$

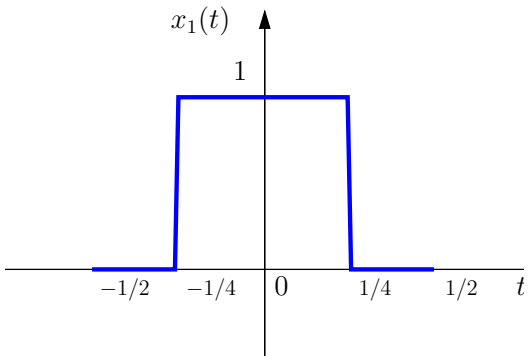


Figure 3: Q2. a)

The signal $x_1(t)$ is periodic with period $T = 1$, $\omega_0 = \frac{2\pi}{T} = 2\pi$ The Fourier series representation of the signal is,

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

where a_k's are the Fourier coefficients given by

$$a_k = \frac{1}{T} \int_{\frac{-T}{2}}^{\frac{T}{2}} x(t) e^{-jk\omega_0 t} dt$$

$$\begin{aligned} a_k &= \int_{-\frac{1}{4}}^{\frac{1}{4}} e^{-j2\pi kt} dt \\ a_k &= \frac{-1}{j2\pi k} \left(e^{-jk\frac{2\pi}{4}} - e^{jk\frac{2\pi}{4}} \right) \\ a_k &= \frac{2}{2\pi k} \left(\sin\left(k\frac{\pi}{2} \right) \right), \text{ for } k \neq 0. \end{aligned}$$

$$a_0 = \int_{-\frac{1}{4}}^{\frac{1}{4}} dt = \frac{1}{2}$$

The magnitude of Fourier coefficient $|a_k| = |\frac{A}{k\pi}\left(\sin\left(k\frac{\pi}{2}\right)\right)|, k \neq 0$ and $|a_0| = \frac{1}{2}$. Phase spectrum of $x_1(t)$ is either zero or π for all n because there is no imaginary part in a_n. The magnitude and phase spectra are shown in Fig. 4.

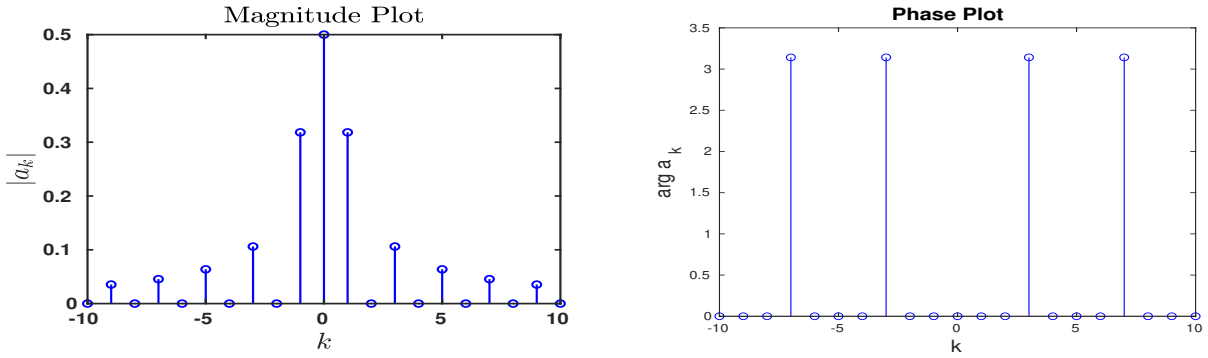


Figure 4: Q2.a) Magnitude and phase spectra of $x_1(t)$ for $T = 1, \omega_0 = 2\pi$.

(b)

$$x_2(t) = \begin{cases} \sin(2\pi t) & 0 \leq t < \frac{1}{2} \\ 0 & \frac{-1}{2} \leq t < 0 \end{cases}$$

The signal $x_2(t)$ is periodic with period $T = 1$.
The Fourier series representation of the signal is,

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$$

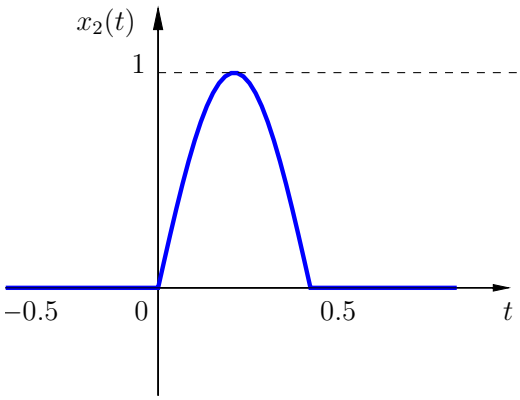


Figure 5: Q2 b)

where $\omega_0 = \frac{2\pi}{T} = 2\pi$, a_n's are the Fourier coefficients given by

$$\begin{aligned} a_0 &= \frac{1}{T}\int_0^{\frac{T}{2}} \sin(\omega_0 t)d(t) \\ a_0 &= \int_0^{\frac{1}{2}} \sin(2\pi t)d(t) \\ a_0 &= -\left(\cos(\pi)-1\right) = \frac{1}{\pi} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{T}\int_{\frac{-T}{2}}^{\frac{T}{2}} \sin(\omega_0 t)e^{-jn\omega_0 t}dt \\ &= \frac{1}{2j}\int_0^{\frac{1}{2}} \left(e^{j2\pi t} - e^{-j2\pi t}\right)e^{-j2\pi nt}dt \\ &= \frac{-1}{2}\left(\frac{e^{j(1-n)\pi}-1}{((1-n)2\pi)} - \frac{e^{-j(1+n)\pi}-1}{-((1+n)2\pi)}\right) \\ &= \frac{-1}{4\pi}\left(\frac{e^{j(1-n)\pi}-1}{(1-n)} + \frac{e^{-j(1+n)\pi}-1}{(1+n)}\right) \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{2\pi}\left(\frac{2}{1-n^2}\right), \quad n = \text{even} \\ a_n &= 0, \quad n = \text{odd},\ n \neq 1,\ n \neq -1 \end{aligned}$$

$$\begin{aligned} a_1 &= \frac{1}{T}\int_0^{\frac{T}{2}} \sin(\omega_0 t)e^{-j\omega_0 t}dt = \frac{-Aj}{4} \\ a_{-1} &= \frac{A}{T_0}\int_0^{\frac{T_0}{2}} \sin(\omega_0 t)e^{j\omega_0 t}dt = \frac{Aj}{4} \end{aligned}$$

Phase spectrum of $x_2(t)$ for $n = 1$ is $-\pi/2$, for $n = -1$ is $\pi/2$. The magnitude and phase spectra are shown in Fig. 6.

3) We have

$$x(t) = \sum_{n=-\infty}^{+\infty} a_n e^{jnt}$$

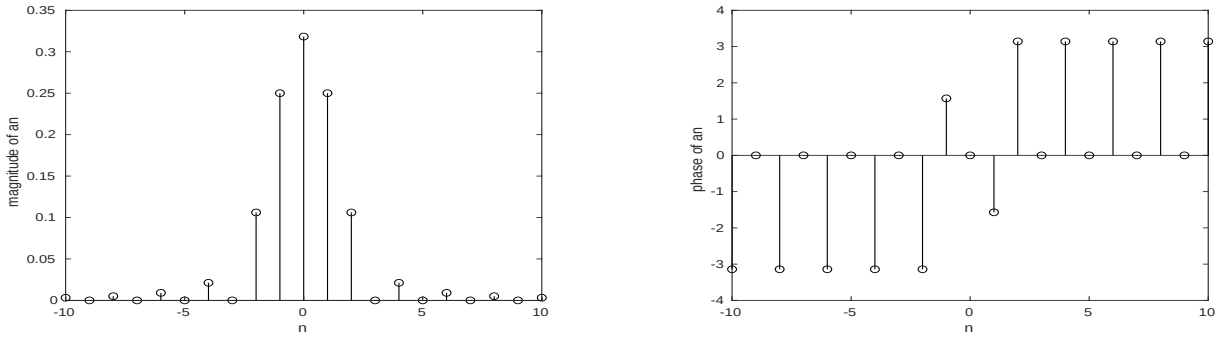


Figure 6: Q2.b) Magnitude and phase spectra of $x_2(t)$ for $T = 1, \omega_0 = 2\pi$.

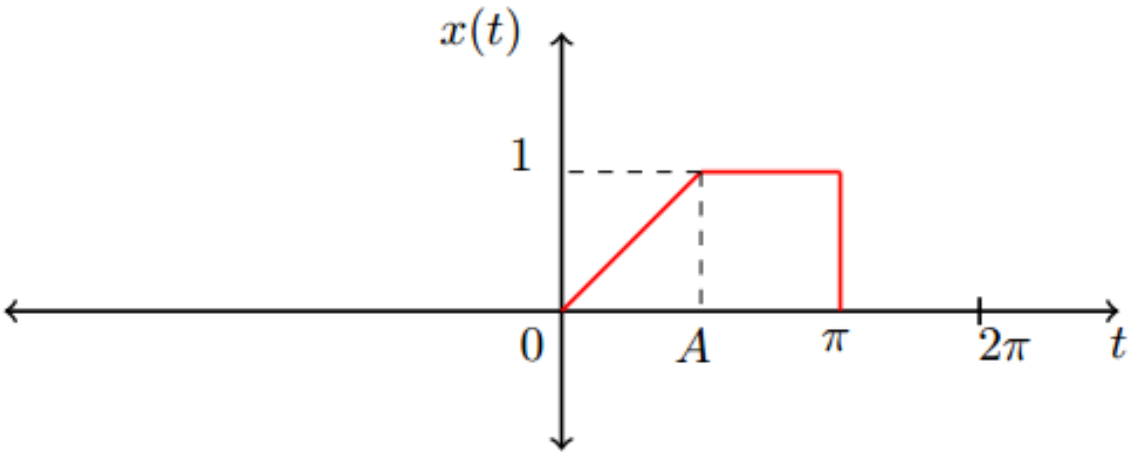


Figure 7: Q3

where

$$a_o = \frac{1}{2\pi}\int_0^{2\pi} x(t)dt$$

$$= \frac{1}{2\pi}\left(\int_0^{A} \frac{t}{A}dt + \int_A^{\pi} dt\right)$$

Therefore,

$$a_o = \frac{1}{2\pi}\left(\pi - \frac{A}{2}\right)$$

and

$$a_n = \frac{1}{2\pi}\int_0^{2\pi} x(t)e^{-jnt}dt$$

$$= \frac{1}{2\pi}\left(\int_0^{A} \frac{t}{A}e^{-jnt}dt + \int_A^{\pi} e^{-jnt}dt\right)$$

Integrating by parts, we have

$$\int_0^A \frac{t}{A} e^{-jnt} dt = j\frac{e^{-jnA}}{n} - \frac{1 - e^{-jnA}}{An^2}$$

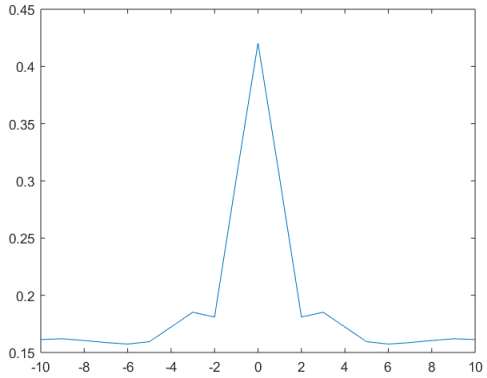
and

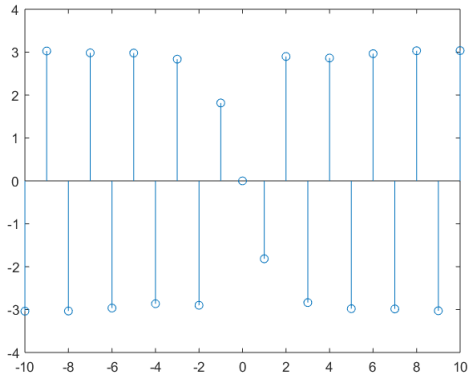
$$\int_A^\pi e^{-jnt} dt = -\frac{j}{n}\left(e^{-jnA} - (-1)^n\right)$$

Therefore,

$$a_n = \frac{1}{2\pi}\left(j\frac{(-1)^n}{n} - \frac{1 - e^{-jnA}}{An^2}\right).$$

Magnitude and phase plot are as follows:





4) $a_k = jk,\ |k| < 3$

$$\begin{aligned} x(t) &= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \\ x(t) &= j\Big(-2e^{j(-2)\frac{2\pi}{4}t} - 1e^{j(-1)\frac{2\pi}{4}t} + 1e^{j\frac{2\pi}{4}t} \\ &\quad + 2e^{j(2)\frac{2\pi}{4}t}\Big) \\ x(t) &= (-1)\left(4\sin(\pi t) + 2\sin(\frac{\pi}{2}t)\right). \end{aligned}$$

5) Given $x(t)$ is a periodic signal with fundamental period T and Fourier series coefficients a_k

(a) Let b_k be the Fourier series coefficient of $x(t - t_0)$ $b_k = \frac{1}{T}\int_T x(t - t_0)e^{-jk\omega_0 t}dt$. The Fourier series representation of the signal is,

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Substitute $t = \tau - t_0$, then

$$\begin{aligned} x(\tau - t_0) &= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0(\tau - t_0)} \\ &= \sum_{k=-\infty}^{\infty} a_k e^{-jk\omega_0 t_0} e^{jk\omega_0 \tau} \end{aligned}$$

Comparing with the synthesis equation, $b_k = a_k e^{-jk\omega_0 t_0}$

(b)

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$\Rightarrow$

$$x(-t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0(-t)} = \sum_{k=-\infty}^{\infty} a_k e^{j(-k)\omega_0 t}$$

Replace k by $-m$ which implies

$$x(-t) = \sum_{m=-\infty}^{\infty} a_{-m} e^{jm\omega_0 t}$$

Thus the Fourier series coefficients of $x(-t)$ is a_{-k}

(c)

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$\Rightarrow$

$$x^*(t) = \sum_{k=-\infty}^{\infty} (a_k e^{jk\omega_0 t})^*$$

$$= \sum_{k=-\infty}^{\infty} a_k^* e^{-jk\omega_0 t}$$

substituting k by $-m$ implies

$$x^*(t) = \sum_{m=-\infty}^{\infty} a_{-m}^* e^{jm\omega_0 t}$$

Thus the Fourier series coefficients of $x^*(t)$ is a_{-k}^*

(d) $x(t - t_0) + x(t + t_0)$

By time shifting property,

$$x(t - t_0) \overset{\mathcal{F}}{\leftrightarrow} a_k e^{-jk\omega_0 t_0}$$

where $\omega_0 = 2\pi/T$. Therefore, Fourier coefficients of $x(t - t_0) + x(t + t_0)$ are

$$a_k e^{-jk\omega_0 t_0} + a_k e^{jk\omega_0 t_0} = 2a_k \cos(k\omega_0 t_0)$$

(e) Even $\{x(t)\} = \dfrac{x(t) + x(-t)}{2}$

$$x(t) \overset{\mathcal{F}}{\leftrightarrow} a_k$$

$$x(-t) \overset{\mathcal{F}}{\leftrightarrow} a_{-k}$$

Therefore, Fourier coefficients of Even $\{x(t)\}$ are $\dfrac{a_k + a_{-k}}{2}$

(f) Real $\{x(t)\} = \dfrac{x(t) + x^*(t)}{2}$

$$x(t) \overset{\mathcal{F}}{\leftrightarrow} a_k$$
$$x^*(t) \overset{\mathcal{F}}{\leftrightarrow} a^*_{-k}$$

Therefore, Fourier coefficients of Real $\{x(t)\}$ are $\dfrac{a_k + a^*_{-k}}{2}$

6) For real-valued signal $a^*_k = a_{-k}$

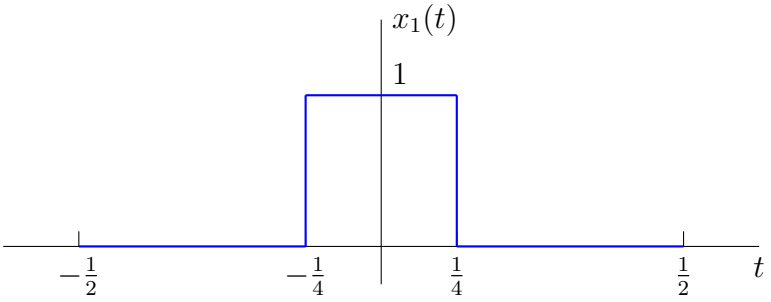
(a) For even signal $a_k = a_{-k}$, which results in $a^*_k = a_k$. That means coefficients are purely real. So there are no sine terms.

(b) For odd signal $a_k = -a_{-k}$, which results in $a^*_k = -a_k$. That means coefficients are purely imaginary. So there are no cosine terms. $a_0 = 0$ as average value of odd signal over period is zero.

7) (a) Fourier series coefficients of $x^*(-t)$ are a^*_k. So for $x(t) = x^*(-t)$, $a_k = a^*_k$. Fourier series coefficients are real valued.

(b) For $x(t) = x(-t)$, Fourier series coefficients are $a_{-k} = a_k$

8) (a) The signal $x_1(t)$ is periodic with period $T_0 = 1$ and $\omega_0 = 2\pi$. Evaluating the Trigonometric Fourier series coefficients,



$$C_0 = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t)dt$$
$$= \int_{-1/4}^{1/4} 1dt$$
$$C_0 = \frac{1}{2}$$

$$C_k = \frac{2}{T_0}\int_{-T_0/2}^{T_0/2} x(t)cos(\omega_0 kt)dt$$
$$= \frac{2}{1}\int_{-1/4}^{1/4} 1 \times cos(2\pi kt)dt$$
$$= 2 \times \frac{1}{2\pi k}(2\sin(2\pi k\frac{1}{4}))$$
$$C_k = \frac{2}{\pi k}\sin(\frac{\pi k}{2})$$

$$C_k = \begin{cases} \frac{1}{2} & k = 0 \\ 0 & k = 2m(even) \\ \frac{2}{\pi k} & k = 4m+1 \\ -\frac{2}{\pi k} & k = 4m+3 \end{cases}$$

$$D_k = \frac{2}{T_0}\int_{-T_0/2}^{T_0/2} x(t)sin(\omega_0 kt)dt$$
$$= \frac{2}{1}\int_{-1/4}^{1/4} 1 \times sin(2\pi kt)dt$$
$$= 2 \times \frac{1}{2\pi k}(-\cos(2\pi t))|_{-1/4}^{1/4}$$
$$D_k = 0$$

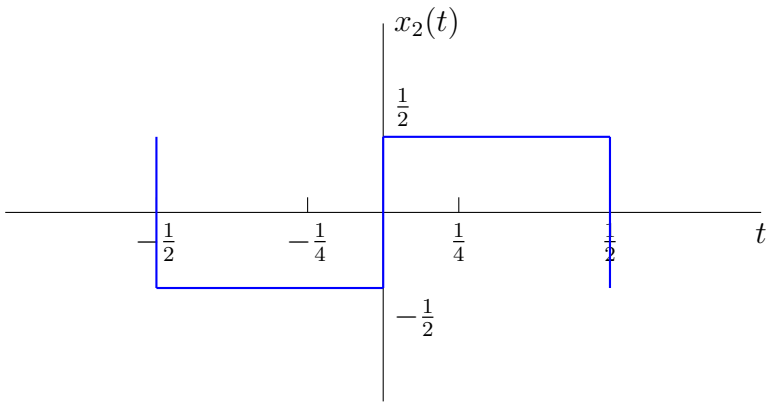
Thus the Fourier series representation of the signal is,

$$x_1(t) = C_0 + \sum_{k=1}^{\infty} C_k cos(\omega_0 kt)$$

Note that the signal is even, therefore the Fourier series representation of the signal has only cosine components.

(b) The signal $x_2(t)$ is periodic with period $T_0 = 1$ and $\omega_0 = 2\pi$. Evaluating the Trigonometric Fourier series coefficients,

$$C_0 = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)dt$$
$$= \int_0^{1/2} \frac{1}{2}dt + \int_{1/2}^{1} -\frac{1}{2}dt$$
$$C_0 = 0$$



$$C_k = \frac{2}{T_0}\int_{-T_0/2}^{T_0/2} x(t)cos(\omega_0 kt)dt$$

$$= 2 \times \int_0^{1/2} \frac{1}{2}cos(2\pi kt)dt + 2 \times \int_{1/2}^{1} -\frac{1}{2}cos(2\pi kt)dt$$

$$= \frac{1}{2\pi k}\left[sin(2\pi kt)|_0^{1/2} - sin(2\pi kt)|_{1/2}^{1}\right]$$

$$C_k = 0 \quad \text{for all k}$$

$$D_k = \frac{2}{T_0}\int_{-T_0/2}^{T_0/2} x(t)sin(\omega_0 kt)dt$$

$$= 2 \times \int_0^{1/2} \frac{1}{2}sin(2\pi kt)dt + \int_{1/2}^{1} -\frac{1}{2}sin(2\pi kt)dt$$

$$= \frac{1}{2\pi k}\left[-cos(2\pi kt)|_0^{\frac{1}{2}} + cos(2\pi kt)|_{\frac{1}{2}}^{1}\right]$$

$$= \frac{1}{\pi k}[-cos(\pi k) + 1]$$

$$= \frac{1}{\pi k}[(-1)^{n+1} + 1]$$

$$D_k = \frac{1}{\pi k}[1 + (-1)^{n+1}]$$

Thus the Fourier series representation of the signal is,

$$x_2(t) = \sum_{k=1}^{\infty} D_k \sin(\omega_0 kt)$$

Note that the signal is odd, therefore the Fourier series representation of the signal has only sine terms (sine is odd).

9) Given signal is a pulse train as shown below

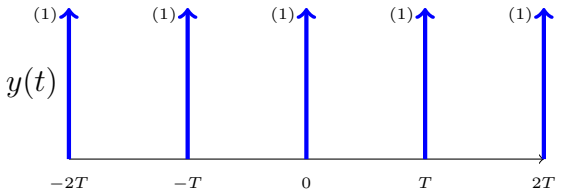


Figure 8: Q7

$$a_n = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} \sum_{k=-\infty}^{\infty} \delta(t-kT)e^{-jn\omega_0 t}dt$$

For the given range, $k = 0$

$$a_n = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} \delta(t)e^{-jn\omega_0 t}dt = \frac{1}{T}$$

10) (a) The plots for signals $x(t)$, $\frac{dx}{dt}$ and $\frac{d^2x}{dt^2}$ are shown in Fig. 10.

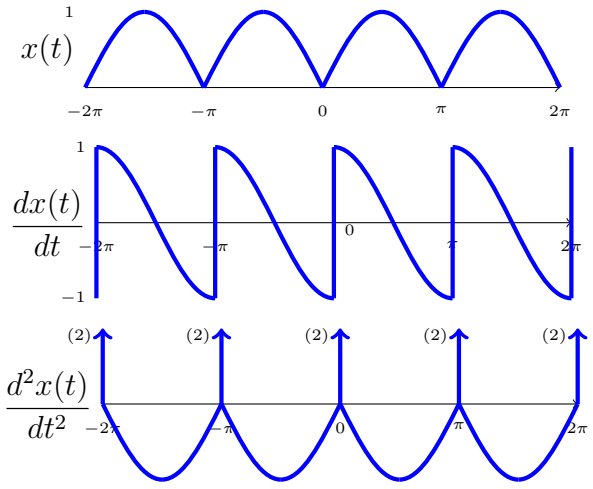


Figure 9: Q8.(a)

(b) The plot for $x(t) + \frac{d^2x(t)}{dt^2}$ is shown in Fig. 11.

$$\begin{aligned} x(t) + \frac{d^2x(t)}{dt^2} &= \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t} \\ &= \sum_{n=-\infty}^{\infty} a_n e^{jn(2)t} \end{aligned}$$

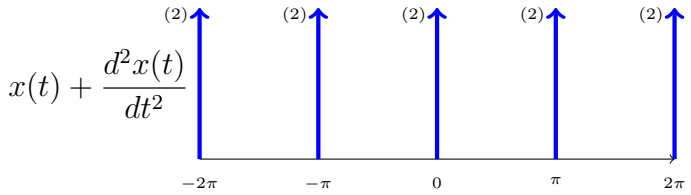


Figure 10: Q8.(b)

where

$$a_n = \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sum_{k=-\infty}^{\infty} 2\delta(t - k\pi) e^{-jn2t} dt$$

Using Q7 and linearity, $a_k = \frac{2}{\pi}$

(c)

$$x(t) \overset{\mathcal{F}}{\leftrightarrow} c_n \qquad \text{and} \qquad \frac{d^2x(t)}{dt^2} \overset{\mathcal{F}}{\leftrightarrow} (jn\omega_0)^2 \, c_n$$

$$x(t) + \frac{d^2x(t)}{dt^2} \overset{\mathcal{F}}{\leftrightarrow} \left(1 + (jn\omega_0)^2\right) c_n$$

$$a_n = \left(1 + (jn\omega_0)^2\right) c_n$$
$$c_n = \frac{a_n}{\left(1 + (jn\omega_0)^2\right)} = \frac{2}{\pi(1 - 4n^2)}$$